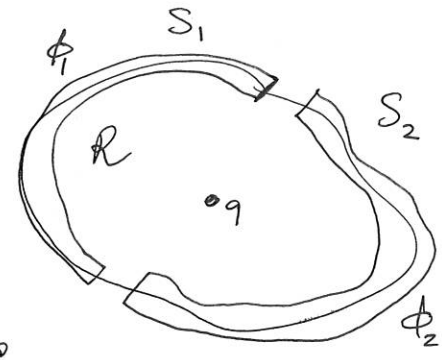
Image charges



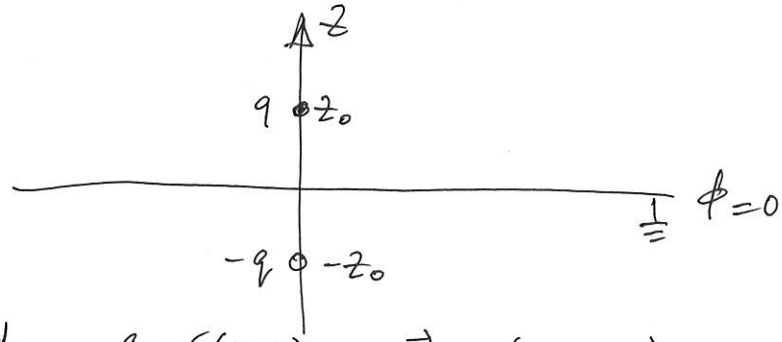
General setting :

charge in some region R.

Potential on surfaces $S_1, S_2 \ldots$ are known to be $\phi_1, \phi_2 \ldots$ Note that some of surfaces may be at ∞.

→ Find electric potential inside R such that boundary conditions are satisfied.

→ The idea: replace S_j with collection of fictitious charges (image charges) such that on S_j the potential takes required values ϕ_j.



(1) $\nabla^2\phi = -\frac{q}{\varepsilon_0}\,\delta(\vec{r}-\vec{r}_0) \qquad \vec{r}_0 = (0,0,z_0).$

BC: $\phi(x,y,0) = 0$

(2) $$\phi(\vec{r}) = \underbrace{\frac{q}{4\pi\varepsilon_0|\vec{r}-\vec{r}_0|}}_{\text{potential of } q} + \underbrace{\frac{-q}{4\pi\varepsilon_0|\vec{r}+\vec{r}_0|}}_{\text{potential of the image charge } -q.}$$

$$\phi(x,y,z) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{\sqrt{x^2+y^2+(z-z_0)^2}} - \frac{1}{\sqrt{x^2+y^2+(z+z_0)^2}}\right)$$

$\Rightarrow$ for $z=0$ $\phi(x,y,0)=0$.

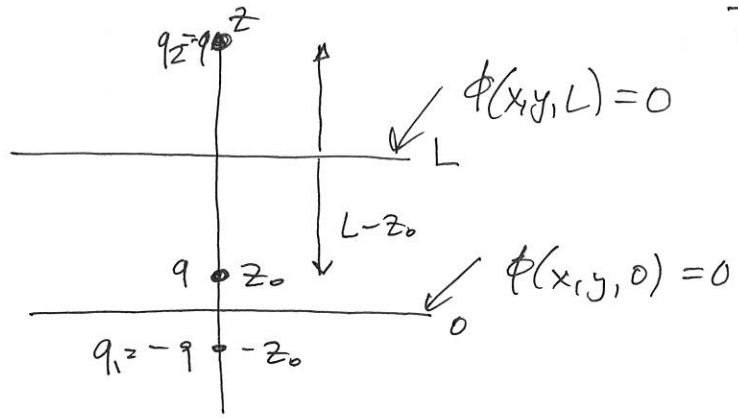
Hence $\phi(\vec{r})$ (2) satisfies BC and solves Poisson's eq (1) everywhere in $z>0$.

Note: eq.(2) does not apply for $z<0$ as there $\nabla^2\phi$ would give $-\frac{(-q)}{\varepsilon_0}\delta(\vec{r}+\vec{r}_0)$ at the location of image charge.

Force of attraction from the plate:

$$\vec{F} = -\hat{z}\,\frac{q^2}{4\pi\varepsilon_0(2z_0)^2}.$$

Two conducting plates.



$$\phi(\vec{r}) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{|\vec{r}-\vec{r}_0|} - \frac{1}{|\vec{r}+\vec{r}_0|}\right)$$ satisfies BC

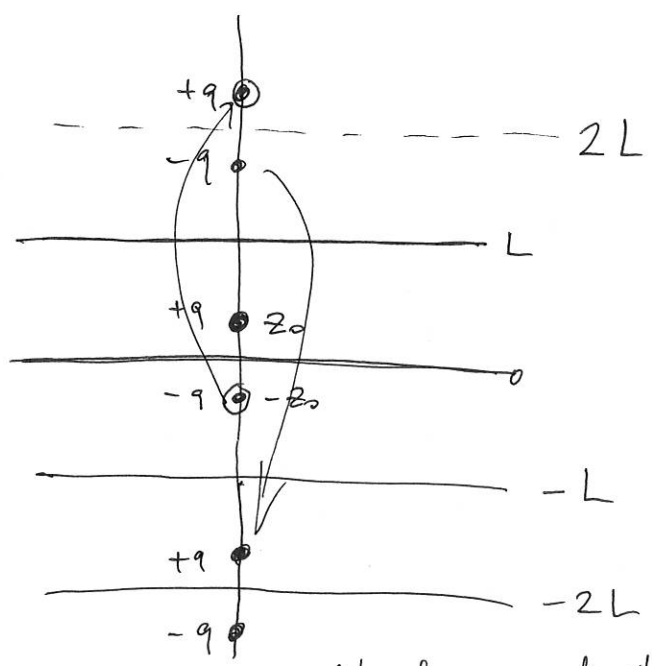
on $z=0$ plate. But at $z=L$

$$\phi(x,y,L) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{\sqrt{x^2+y^2+(L-z_0)^2}} - \frac{1}{\sqrt{x^2+y^2+(L+z_0)^2}}\right)$$

it does not!

Need another image q_2 at $z = L+L-z_0$

and then image of q_1 with respect to L-plane.



Need an infinite number of images!

Change gears - do Fourier transform.

$$\nabla^2 \phi(\vec{r}) = -\frac{\rho(\vec{r})}{\varepsilon_0}$$

$$\underbrace{\int d^3\vec{r}\; e^{-i\vec{k}\cdot\vec{r}}\, \nabla^2\phi}_{\text{integrate by parts}} = -\frac{1}{\varepsilon_0} \underbrace{\int d^3\vec{r}\; e^{-i\vec{k}\cdot\vec{r}}\, \rho(\vec{r})}_{\rho(\vec{k})}$$

$$\|$$

$$\int d^3\vec{r}\; \phi(\vec{r})\, \nabla^2 e^{-i\vec{k}\cdot\vec{r}} = -k^2 \int d^3\vec{r}\; \phi(\vec{r})\, e^{-i\vec{k}\cdot\vec{r}} = -k^2 \phi(\vec{k})$$

$\Rightarrow$ obtain $\quad k^2 \phi(\vec{k}) = \dfrac{\rho(\vec{k})}{\varepsilon_0}$

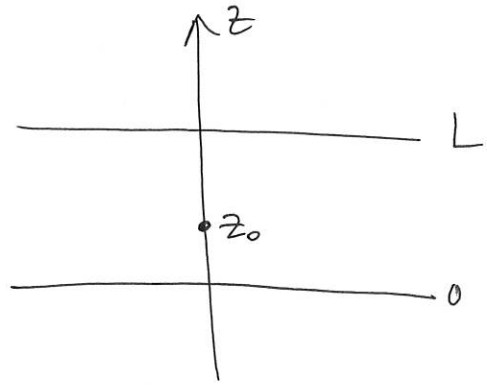
For a point charge $\rho(\vec{r}) = q(\vec{r} - \vec{r}_0)$

$$\Rightarrow \rho(\vec{k}) = q\, e^{-i\vec{k}\cdot\vec{r}_0}$$

$$\Rightarrow \phi(\vec{k}) = \frac{q}{\varepsilon_0 k^2}\, e^{-i\vec{k}\cdot\vec{r}_0}$$

And now we just need to Fourier back to $\vec{r}$ - space:

$$\phi(\vec{r}) = \int \frac{d^3\vec{k}}{(2\pi)^3}\, \phi(\vec{k})\, e^{i\vec{k}\cdot\vec{r}}$$



$$\nabla^2 \phi(\vec{r}) = -\frac{q}{\varepsilon_0}\,\delta(x)\,\delta(y)\,\delta(z-z_0) \quad (1)$$

let us FT with respect to transverse coordinates

$\vec{\rho} = (x, y, 0)$

$$\phi(\vec{r}) = \phi(x,y,z) = \phi(\vec{\rho}, z) = \int \frac{d^2\vec{k}}{(2\pi)^2}\, e^{-i\vec{k}\cdot\vec{\rho}}\, \phi(\vec{k}, z)$$

$$\Rightarrow \nabla^2 \phi(\vec{r}) = \frac{\partial^2}{\partial z^2}\phi(\vec{k}, z) - k^2\,\phi(\vec{k}, z)$$

We obtain:

$$\frac{\partial^2 \phi(k,z)}{\partial z^2} - k^2\,\phi(k,z) = -\frac{q}{\varepsilon_0}\,\delta(z-z_0) \quad (2)$$

BC: $\phi(x,y,0) = \phi(x,y,L) = 0 \Rightarrow \phi(\vec{k},0) = \phi(\vec{k},L) = 0.$

Now: $0 < z < z_0 \Rightarrow \delta(z - z_0) = 0$

$$\frac{\partial^2 \phi}{\partial z^2} = k^2 \phi \Rightarrow \phi(k,z) = a_1 e^{kz} + a_2 e^{-kz}$$

But we need $\phi(k,0) = 0 \Rightarrow a_1 + a_2 = 0$

$$\Rightarrow \phi(k,z) = a_1\left(e^{kz} - e^{-kz}\right) = 2a_1\, sh(kz)$$

Next $z_0 < z < d \Rightarrow \phi(k,z) = 2b_1 \, sh\, k(L-z)$

satisfies $\frac{\partial^2 \phi}{\partial z^2} = k^2 \phi$ <u>and</u> $\phi(\vec{k}, L) = 0$.

In addition, $\phi(\vec{k}, z)$ must be continuous at $z = z_0$

$$2a_1 \, sh(kz_0) = 2b_1 \, sh\, k(L-z_0)$$

$\Rightarrow$ let $2a_1 = C \, sh\, k(L-z_0)$

$2b_1 = C \, sh(kz_0)$

$$\Rightarrow \phi(\vec{k}, z) = \begin{cases} C \, sh\, k(L-z_0) \, sh(kz) & 0 < z < z_0 \\ C \, sh(kz_0) \, sh\, k(L-z) & z_0 < z < L \end{cases}$$

To find C: integrate (2) over $z \in (z_0 - \delta, z_0 + \delta)$

$$\int_{z_0-\delta}^{z_0+\delta} dz \, \frac{\partial^2 \phi}{\partial z^2} - \int_{z_0-\delta}^{z_0+\delta} dz \, k^2 \phi = -\frac{q}{\varepsilon_0} \int_{z_0-\delta}^{z_0+\delta} dz \, \delta(z-z_0)$$

Take limit $\delta \to +0$:

$$\Rightarrow \left.\frac{\partial \phi}{\partial z}\right|_{z=z_0+0} - \left.\frac{\partial \phi}{\partial z}\right|_{z_0-0} = -\frac{q}{\varepsilon_0}$$

$$-Ck \, sh(kz_0) \, sh\, k(L-z_0) - C\cancel{k} \, sh\, k(L-z_0) \, ch\, kz_0 = -\frac{q}{\varepsilon_0}$$

$$C\left\{sh(kz_0) \, ch[k(L-z_0)] + sh(kz_0) \, sh[k(L-z_0)]\right\} = \frac{q}{\varepsilon_0 k}$$

identity!

$$C\left\{ sh[kz_0 + k(L-z_0)] \right\} = \frac{q}{\varepsilon_0 k}$$

$$\Rightarrow C = \frac{q}{\varepsilon_0 k \, sh(kL)}$$

What is energy of this configuration?

$$\mathcal{E} = \frac{1}{2} \sum q_i \Phi(\vec{r}_i) = \frac{1}{2} q \, \phi(z_0)$$

$$\phi(\vec{\rho}, z_0) = \int \frac{d^2\vec{k}}{(2\pi)^2} e^{-i\vec{k}\cdot\vec{\rho}} \phi(\vec{k}, z)$$

$$\Rightarrow \phi(\rho=0, z_0) = \int \frac{d^2\vec{k}}{(2\pi)^2} \phi(\vec{k}, z_0) =$$

$$= \int \frac{d\vec{k}}{(2\pi)^2} \frac{q}{\varepsilon_0 k \, sh(kL)} sh(kz_0) sh[k(L-z_0)]$$

This integral diverges for large k — this is just energy of self-interaction of the point charge with itself.

But the force acting on the charge

$$F = -\frac{\partial \mathcal{E}}{\partial z_0} \text{ is finite!}$$

$$F = -\frac{q^2}{2\varepsilon_0} \int \frac{d^2\vec{k}}{(2\pi)^2} \frac{\partial}{\partial z_0} \left(\frac{sh(kz_0) \, sh \, k(L-z_0)}{k \, sh(kL)} \right)$$

$$\frac{\partial}{\partial z_0}\left(\quad\right) = \frac{1}{k}\,\frac{k\,ch(kz_0)\,sh\,k(L-z_0) - k\,sh(kz_0)\,ch\,k(L-z_0)}{sh(kL)}$$

$$\cancel{-\frac{1}{k}\,\frac{sh(kz_0)\,sh\,k(L-z_0)}{sh^2(kL)}} = \frac{k}{k}\,\frac{sh[kz_0 - k(L-z_0)]}{sh(kL)} =$$

$$= \frac{sh\,k(2z_0-L)}{sh(kL)}$$

$$F = -\frac{q^2}{2\varepsilon_0}\int_0^\infty \frac{dk\cdot k\cdot 2\pi}{(2\pi)^2}\,\frac{sh\,k(2z_0-L)}{sh(kL)}$$

$$= +\frac{q^2}{4\pi\varepsilon_0}\int_0^\infty dk\,k\,\frac{sh\,k(L-2z_0)}{sh(kL)}$$

Consider limits: (1) $z_0 \ll L$

$$sh\,k(L-2z_0) \approx \frac{1}{2}e^{k(L-2z_0)}\ ;\ sh(kL) \simeq \frac{1}{2}e^{kL}$$

$$F \approx \frac{q^2}{4\pi\varepsilon_0}\int_0^\infty dk\,k\,e^{-2z_0 k} = \frac{q^2}{4\pi\varepsilon_0}\,\frac{1}{(2z_0)^2}$$

This is force acting on q due to the nearest to it conducting plate at $z=0$!

(2) let $z_0 = \frac{1}{2}L + \Delta z \Rightarrow L - 2z_0 = -2\Delta z$

$$F = -\frac{q^2}{4\pi\varepsilon_0}\int_0^\infty dk\,k\,\frac{sh(2\Delta z\,k)}{sh(kL)} \approx -\frac{q^2\Delta z}{2\pi\varepsilon_0}\int_0^\infty dk\,\frac{k^2}{sh(kL)}$$

$$\int_0^\infty \frac{dk\,k^2}{sh(kL)} = \frac{1}{L^3}\int \frac{dx\,x^2}{sh(x)} = \frac{1}{L^3}\cdot \text{number.}$$

$\hookrightarrow \approx 4.2$